



Software Development for Mobile Devices

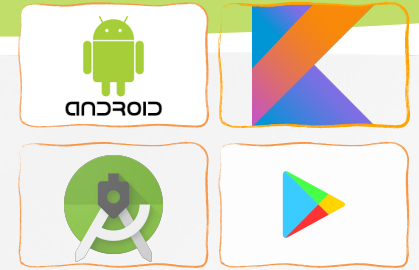
Nguyen Le Hoang Dung
[*nlhdung@fit.hcmus.edu.vn*](mailto:nlhdung@fit.hcmus.edu.vn)

Content

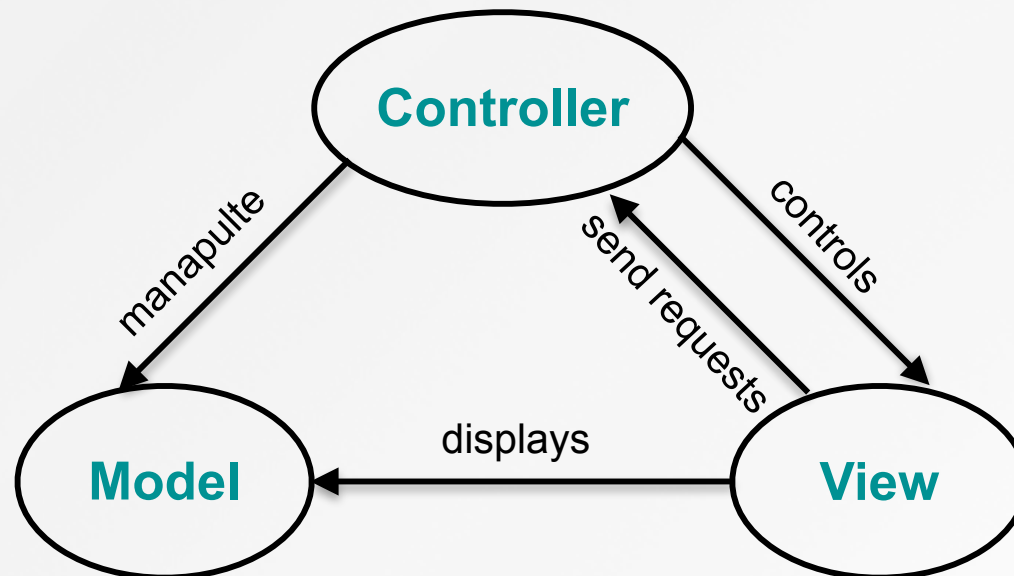
» MVC

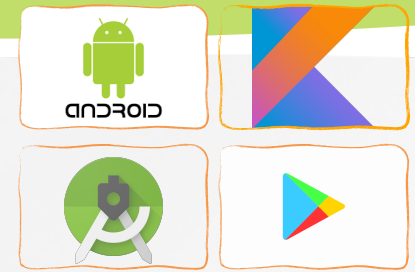


The Model-View-Controller Pattern (MVC)



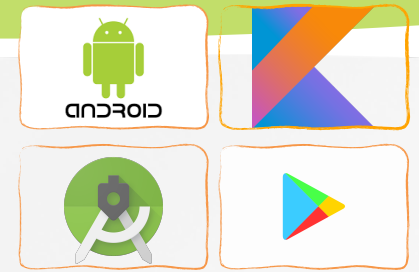
» The Model-View-Controller (MVC) is an important software design pattern whose main goal is to separate the (1) user interface, (2) business, and (3) input logic.





The MVC Pattern (cont.)

- » Model: Consists of the Java code and API objects
 - Represent the business problem.
 - Manage the behavior and data of the application.
- » View: Set of screens the user sees and interacts with.
- » Controller: Implemented through the Android OS
 - Responsible for interpretation of the user and system inputs. Input may come from a variety of sources such as the trackball, keyboard, touch-screen, GPS chip, proximity sensor, accelerometer, etc...
 - Tells the Model and/or the View (usually through callbacks and registered listeners) to change as appropriate.

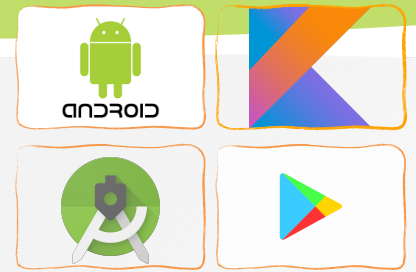


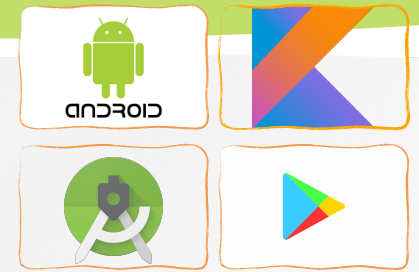
The MVC Pattern (cont.)

We should be aware of

- » Inputs could be sent to the application from various physical/logical components. Reacting to those signals is typically handled by callback methods.
- » Moving to states in the lifecycle is tied to logic in the model. For example, if forced to Pause you may want to save uncommitted data.
- » A notification mechanism is used to inform the user of important events happening outside the current application (such as arrival of a text message or email, low battery, fluctuations of the stock market, etc) and consequently choose how to proceed.
- » Views (Android graphical interfaces) are usually implemented as XML files (although they could also be dynamically created from Java code).

Q&A





Reference

- » <https://kotlinlang.org/docs/reference/>
- » Peter Späth, Learn Kotlin for Android Development, 2019
- » Ted Hagos, Learn Android Studio 3 with Kotlin, 2018
- » Peter Späth, Pro Android with Kotlin, 2019
- » Milos Vasic, Mastering Android Development with Kotlin, 2017
- » <https://developer.android.com/guide/>